OPLOGS-04: Buckets & Data Governance

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Strategic Storage Management for OpenPipeline Logs

This notebook covers Grail bucket architecture, retention policies, routing configuration, access control, and cost optimization strategies.

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Prerequisites

- ✅ Access to a Dynatrace environment with log data
- ✅ Understanding of OpenPipeline basics (OPLOGS-01 to OPLOGS-03)
- ✅ Admin access for bucket configuration (optional)

1. Discovering Your Buckets

Start with what already exists: which buckets hold log data, how their volume moves, and which log levels land in each.

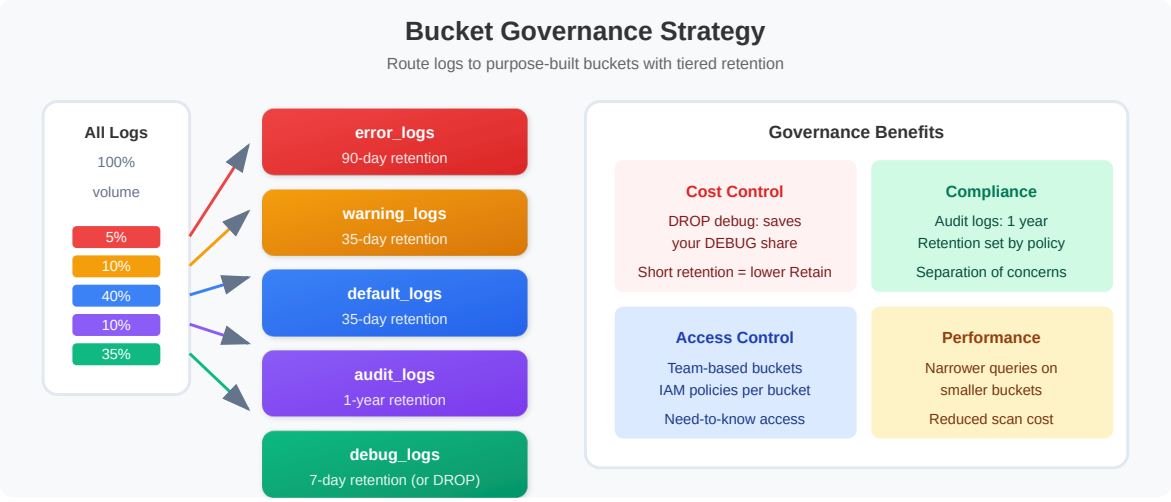
```
// Discover all buckets with log data
fetch logs, from: now() - 24h
| summarize {log_count = count()}, by: {dt.system.bucket}
| sort log_count desc
```

```
// Bucket volume analysis over time
fetch logs, from: now() - 7d
| makeTimeseries {count = count()}, by: {dt.system.bucket}, interval: 24h
| limit 100
```

```
// Bucket composition by log level
fetch logs, from: now() - 24h
| summarize {count = count()}, by: {dt.system.bucket, loglevel}
| sort dt.system.bucket asc, count desc
```

2. Strategic Bucket Design

Design buckets based on **access patterns**, **retention requirements**, and **cost optimization**.



Recommended Bucket Strategy

Bucket Name	Purpose	Retention	Access	Data Types
`default_logs`	General purpose	35 days	All teams	Standard logs
`debug_logs`	Development troubleshooting	7 days	Dev teams	DEBUG, TRACE
`error_logs`	Incident investigation	90 days	SRE/Ops	ERROR, FATAL
`audit_logs`	Compliance	365 days	Audit team	Auth, access
`security_logs`	Security operations	180 days	SecOps	Security events
`archive_logs`	Long-term storage	730 days	Compliance	Critical events

Bucket Naming Convention

```
&lt;purpose&gt;_logs
&lt;team&gt;_&lt;purpose&gt;_logs
&lt;environment&gt;_&lt;purpose&gt;_logs
```

Examples:

- `platform_debug_logs`
- `prod_security_logs`
- `payment_audit_logs`

```
// Analyze current log distribution for bucket planning
// status == "ERROR" covers every error-class level (ERROR, SEVERE, CRITICAL,
FATAL, …);
// loglevel == "ERROR" alone misses SEVERE and the rest
fetch logs, from: now() - 24h
| summarize {
    total = count(),
    debug = countIf(loglevel == "DEBUG" OR loglevel == "TRACE"),
    info = countIf(loglevel == "INFO" OR loglevel == "NOTICE"),
    warn = countIf(loglevel == "WARN"),
    error = countIf(status == "ERROR")
  }
| fieldsAdd debug_pct = round((debug * 100.0) / total, decimals: 1)
| fieldsAdd info_pct = round((info * 100.0) / total, decimals: 1)
| fieldsAdd warn_pct = round((warn * 100.0) / total, decimals: 1)
| fieldsAdd error_pct = round((error * 100.0) / total, decimals: 1)
```

```
// Identify audit/security logs for specialized bucket
fetch logs, from: now() - 24h
| filter contains(content, "auth")
        OR contains(content, "login")
        OR contains(content, "logout")
        OR contains(content, "permission")
        OR contains(content, "access")
| summarize {count = count()}, by: {k8s.namespace.name, loglevel}
| sort count desc
| limit 20
```

```
// Volume by source - helps determine routing needs
fetch logs, from: now() - 24h
| summarize {count = count()}, by: {dt.openpipeline.source, dt.system.bucket}
| sort count desc
```

3. Retention Policies

Retention Strategy Matrix

Starting points from community practice — your audit, legal and investigation needs set the real numbers:

Data Type	Min Retention	Recommended	Max	Rationale
DEBUG/TRACE	3 days	7 days	14 days	High volume, temporary
INFO	7 days	35 days	90 days	Standard operations
WARN	14 days	60 days	180 days	Trend analysis
ERROR	30 days	90 days	365 days	Incident investigation
Audit	365 days	365 days	730 days	Compliance
Security	90 days	180 days	365 days	SIEM/forensics

Retention Cost Impact

Under DPS, Log Management & Analytics has separate **Ingest & Process**, **Retain** and **Query** dimensions (usage-based model; *Retain with Included Queries* is the alternative model). Retention changes only the **Retain** dimension:

```
Retain consumption per day = GiB stored × Retain GiB-day price
GiB stored (steady state)  ≈ daily volume × retention days

Example (DEBUG logs):
- 100 GB/day × 35 days = 3,500 GB stored
- 100 GB/day ×  7 days =   700 GB stored
- Result: 80% fewer DEBUG GiB-days retained
- Ingest &amp;amp; Process is billed per GiB ingested, so it does not change
```

> **Sources:** Log Analytics overview (DT docs) — *"In the usage-based model, Log - Retain and Log - Query are charged separately"*, Log Management & Analytics - Retain (DT docs) — *"consumption per day = (volume of uncompressed logs stored in GiB) x (GiB-day price for Retain on your rate card)"*, Log Management & Analytics - Ingest & Process (DT docs) — *"It is calculated per gibibyte (GiB)."*

```
// Estimate storage by log level
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {
    log_count = count(),
    total_mb = round(sum(content_bytes) / 1048576.0, decimals: 2),
    avg_bytes = round(avg(content_bytes), decimals: 0)
  }, by: {loglevel}
| sort total_mb desc
```

```
// Retained DEBUG/TRACE volume at 35 vs 7 days (steady state)
// Retain is billed per GiB-day stored; Ingest &amp;amp; Process (per GiB
ingested) does not change with retention
fetch logs, from: now() - 24h
| filter loglevel == "DEBUG" OR loglevel == "TRACE"
| fieldsAdd content_bytes = stringLength(content)
| summarize {daily_mb = sum(content_bytes) / 1048576.0}
| fieldsAdd daily_ingest_gb = round(daily_mb / 1024, decimals: 2)
| fieldsAdd stored_35d_gb = round(daily_mb * 35 / 1024, decimals: 2)
| fieldsAdd stored_7d_gb = round(daily_mb * 7 / 1024, decimals: 2)
| fieldsAdd retain_gib_days_avoided_per_day = round(daily_mb * (35 - 7) /
1024, decimals: 2)
| fieldsAdd daily_mb = round(daily_mb, decimals: 2)
```

```
// Retention planning: volume by bucket and loglevel
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {
    log_count = count(),
    daily_mb = round(sum(content_bytes) / 1048576.0, decimals: 2)
  }, by: {dt.system.bucket, loglevel}
| sort dt.system.bucket asc, daily_mb desc
```

4. Bucket Routing Configuration

A **route** sends a record to a **pipeline**; the **bucket** is chosen inside that pipeline by **Bucket assignment** processors in the Bucket assignment stage. There is no route-level bucket setting.

Bucket Assignment Processors

#	Matching condition	Bucket
1	`loglevel == "DEBUG" OR loglevel == "TRACE"`	`debug_logs`
2	`status == "ERROR"`	`error_logs`
3	`matchesValue(content, "*audit*") OR matchesValue(k8s.namespace.name, "*security*")`	`audit_logs`

`contains()` is not enabled in OpenPipeline matching conditions, hence `matchesValue`. `status == "ERROR"` covers SEVERE, CRITICAL and FATAL as well as ERROR.

Assignment Priority

1. Bucket assignment is **first match only** — the first processor whose condition matches decides
2. Order processors from specific to general
3. A last processor with condition `true` makes the fallback explicit; in community practice, records that match none land in the default bucket

> **Sources:** Processing in OpenPipeline (DT docs) — Bucket assignment stage: *"Assign records to the best-fit bucket."*, executed *first match only*.

```
// Preview routing decision logic over the whole hour (the summarize bounds the output;
// a limit before it lets the first records decide the mix and hides whole branches)
fetch logs, from: now() - 1h
| fieldsAdd target_bucket = if(loglevel == "DEBUG" OR loglevel == "TRACE", "debug_logs",
                            else: if(status == "ERROR", "error_logs",
                            else: if(contains(content, "audit") OR contains(content, "security"), "audit_logs",
                            else: "default_logs")))
| summarize {count = count()}, by: {target_bucket, loglevel}
| sort target_bucket asc, count desc
```

```
// Validate current routing effectiveness
fetch logs, from: now() - 24h
| summarize {
    total = count(),
    in_default = countIf(dt.system.bucket == "default_logs")
  }, by: {loglevel}
| fieldsAdd default_pct = round((in_default * 100.0) / total, decimals: 1)
| sort default_pct desc
```

```
// Namespace-based routing analysis
fetch logs, from: now() - 24h
| filter isNotNull(k8s.namespace.name)
| fieldsAdd suggested_bucket = if(contains(k8s.namespace.name, "prod"),
"prod_logs",
                               else: if(contains(k8s.namespace.name,
"staging"), "staging_logs",
                               else: "dev_logs"))
| summarize {count = count()}, by: {k8s.namespace.name, suggested_bucket}
| sort count desc
| limit 20
```

5. Access Control & Data Governance

Bucket-Level Access Control

Bucket	Read Access	Write Access	Use Case
`default_logs`	All authenticated	OpenPipeline	General access
`debug_logs`	Dev teams	OpenPipeline	Development
`audit_logs`	Audit team only	OpenPipeline	Compliance
`security_logs`	SecOps only	OpenPipeline	Security ops

IAM Policy Integration

Configure in **Account Management** → **Identity & access management** → **Policy management**. Policies are written in the policy statement language, not JSON, and bucket access needs **two** statements — read on the bucket, plus the table permission:

```
ALLOW storage:buckets:read WHERE storage:bucket-name="audit_logs";
ALLOW storage:logs:read WHERE storage:bucket-name="audit_logs";
```

> **Sources:** Permissions in Grail (DT docs) — *"All bucket permissions need to start with storage:buckets:read"*, Manage IAM policies (DT docs) — *"Go to Account Management > Identity & access management > Policy management."*

```
// Identify sensitive data patterns for access control planning
fetch logs, from: now() - 24h
| filter contains(content, "password")
        OR contains(content, "secret")
        OR contains(content, "token")
        OR contains(content, "api_key")
        OR contains(content, "credential")
| summarize {count = count()}, by: {dt.system.bucket, k8s.namespace.name}
| sort count desc
| limit 15
```

```
// Data governance audit: bucket access patterns
fetch logs, from: now() - 24h
| summarize {
    total_logs = count(),
    unique_sources = countDistinct(dt.openpipeline.source)
  }, by: {dt.system.bucket}
| sort total_logs desc
```

6. Query Optimization with Buckets

Target Specific Buckets

Querying specific buckets **reduces scan costs** and **improves performance**.

✅ **Efficient - targets specific bucket:**

```
fetch logs, bucket: {"error_logs"}, from: now() - 7d
| filter status == "ERROR"
| limit 100
```

❌ **Inefficient - scans all buckets:**

```
fetch logs, from: now() - 7d
| filter status == "ERROR"
| limit 100
```

Multi-Bucket Queries

```
// Query multiple specific buckets
fetch logs, bucket: {"prod_logs", "error_logs"}, from: now() - 24h
| filter status == "ERROR"
```

```
// Query specific bucket (efficient)
fetch logs, bucket: {"default_logs"}, from: now() - 1h
| summarize {count = count()}, by: {loglevel}
| sort count desc
```

```
// Compare bucket query efficiency
fetch logs, from: now() - 1h
| summarize {
    total_logs = count(),
    buckets_scanned = countDistinct(dt.system.bucket)
  }
```

```
// Bucket-aware dashboarding: hourly volume by bucket
fetch logs, from: now() - 24h
| makeTimeseries {count = count()}, by: {dt.system.bucket}, interval: 1h
| limit 100
```

7. Cost Optimization Strategies

Storage Cost Reduction Tactics

Strategy	Effect	Implementation
Shorten DEBUG retention	Retained DEBUG volume falls with retention (7 of 35 days ≈ 80% less)	Bucket assignment to a 7-day bucket

Strategy	Effect	Implementation
Drop health check logs	Saves your health-check share — measure it first (OPLOGS-03 §7)	Drop record processor
Trim long content	Depends on how much of each record you can remove	DQL processor (`fieldsRemove` , or a shortened `content`)

ROI Calculation

```
Current State:
- 100 GB/day total logs
- 35-day retention all buckets
- Total storage: 3,500 GB

Optimized State:
- 60 GB/day INFO/WARN (35d) = 2,100 GB
- 30 GB/day DEBUG (7d) = 210 GB
- 10 GB/day ERROR (90d) = 900 GB
- Total storage: 3,210 GB

Stored volume: ~8% lower (Retain); ingest unchanged
```

```
// Stored volume per log level: everything at 35 days vs tiered retention
// delta_gb &gt; 0: less stored; delta_gb &lt; 0: more stored (90-day error
retention adds Retain cost)
// Rounding happens only on the outputs, so the products are not distorted
fetch logs, from: now() - 24h
| fieldsAdd content_bytes = stringLength(content)
| summarize {daily_gb = sum(content_bytes) / 1073741824.0, log_count =
count()}, by: {loglevel}
| fieldsAdd tiered_retention_days = if(loglevel == "DEBUG" OR loglevel ==
"TRACE", 7,
                                    else: if(in(loglevel, {"ERROR", "SEVERE",
"CRITICAL", "FATAL", "ALERT", "EMERGENCY"}), 90,
                                    else: 35))
| fieldsAdd stored_35d_gb = round(daily_gb * 35, decimals: 1)
| fieldsAdd stored_tiered_gb = round(daily_gb * tiered_retention_days,
decimals: 1)
| fieldsAdd delta_gb = round(daily_gb * (35 - tiered_retention_days),
decimals: 1)
| fieldsAdd daily_gb = round(daily_gb, decimals: 3)
| sort delta_gb desc
```

```
// Identify high-volume low-value logs
fetch logs, from: now() - 24h
| filter loglevel == "DEBUG" OR loglevel == "INFO"
| fieldsAdd content_bytes = stringLength(content)
| summarize {
    log_count = count(),
    total_mb = round(sum(content_bytes) / 1048576.0, decimals: 2)
  }, by: {k8s.namespace.name, k8s.workload.name}
| sort total_mb desc
| limit 20
```

8. Bucket Management Best Practices

✅ Do's

1. **Start with clear bucket strategy** before configuring pipelines
2. **Name buckets descriptively** with purpose and team
3. **Set retention based on data value**, not convenience
4. **Review bucket distribution** weekly
5. **Query specific buckets** for performance
6. **Document bucket purposes** and access policies

❌ Don'ts

1. **Don't put everything in default bucket** - defeats purpose
2. **Don't set uniform retention** - wastes storage on low-value data
3. **Don't ignore bucket metrics** - monitor distribution
4. **Don't create too many buckets** - complexity overhead
5. **Don't skip access controls** - security risk

```
// Weekly bucket health check query
// error_pct uses status == "ERROR", which includes SEVERE, CRITICAL, FATAL, …
fetch logs, from: now() - 7d
| summarize {
    total_logs = count(),
    unique_sources = countDistinct(dt.openpipeline.source),
    debug_pct = round((countIf(loglevel == "DEBUG") * 100.0) / count(),
decimals: 1),
    error_pct = round((countIf(status == "ERROR") * 100.0) / count(),
decimals: 1)
  }, by: {dt.system.bucket}
| sort total_logs desc
```

```
// Bucket routing effectiveness
fetch logs, from: now() - 24h
| summarize {count = count()}, by: {dt.system.bucket,
dt.openpipeline.pipelines}
| sort count desc
```

📝 Summary

In this notebook, you learned:

✅ **Bucket Architecture** - How Grail organizes log storage
✅ **Strategic Design** - Creating purpose-driven buckets
✅ **Retention Policies** - Optimizing data lifecycle costs
✅ **Routing Rules** - Directing logs to appropriate storage
✅ **Access Control** - Implementing data governance
✅ **Query Optimization** - Targeting buckets for efficiency

Key Takeaway

> **Buckets are foundational.** Design your bucket strategy first - it impacts security, cost, and every query you'll write.

➡️ Next Steps

Continue to **OPLOGS-05: Querying & Parsing** to learn advanced DQL techniques for log analysis.

📚 References

- Use Grail buckets to partition data (DT docs)
- Configure data storage and retention for logs (DT docs)
- Data retention periods (DT docs)
- Working with policies (DT docs)
- Permissions in Grail (DT docs)
- Log Management & Analytics - Retain (DT docs)
- Log Management & Analytics - Ingest & Process (DT docs)

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